Lecture 16: The Physics of Control
(Delayed Neutrons and the Inhour Equation)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Reading Assignment

Lamarsh & Baratta (4th Edition):

- **Section 7.2** Reactor Kinetics (Prompt/Delayed Neutrons, Inhour Equation, Prompt Critical).

Recap: The Problem with Prompt Neutrons

In Lecture 15, we saw that if all neutrons in a reactor were prompt neutrons, the time scale of the reaction (the Reactor Period T) would be on the order of milliseconds.

$$T \approx \frac{l}{\rho} \quad \text{where} \quad \rho = \frac{k-1}{k}$$

If $l = 10^{-4}$ s and $\rho = 0.001$, the power rises by a factor of 22,000 in one second. This is uncontrollable. Today, we mathematically prove how **Delayed Neutrons** save us.

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1 The Point Kinetics Equations

To account for delayed neutrons, we must modify our neutron balance equation. We split the neutron source into two terms:

1. **Prompt Neutrons:** $(1-\beta)$ fraction of fission neutrons appear instantly.
2. **Delayed Neutrons:** β fraction of fission neutrons appear later, coming from the decay of Precursors (C).

We now need two coupled differential equations: one for the Neutrons (n) and one for the Precursors (C).

1.1 1. The Neutron Equation

$$\frac{dn}{dt} = \underbrace{\frac{k(1-\beta)n}{l}}_{\text{Prompt Generation}} + \underbrace{\lambda C}_{\text{Delayed Source}} - \underbrace{\frac{n}{l}}_{\text{Loss}}$$

Here, λ is the decay constant of the precursor (e.g., ^{87}Br).

1.2 2. The Precursor Equation

The precursors are created by fission and lost by radioactive decay.

$$\frac{dC}{dt} = \underbrace{\frac{k\beta n}{l}}_{\text{Production}} - \underbrace{\lambda C}_{\text{Decay}}$$

2 The Solution: The Inhour Equation

We look for solutions of the form $n(t) = n_0 e^{t/T}$ (exponential growth with period T). Substituting this ansatz into the coupled system involves some algebra (omitted here for brevity, see Lamarsh Ch. 7), but the result is the fundamental equation of reactor control:

$$\rho = \frac{l}{Tk_{eff}} + \frac{\beta_{eff}}{1 + \lambda T} \tag{1}$$

The Inhour Equation (Simplified One-Group Form)

Historical Note: You might wonder who "Dr. Inhour" was. There is no such person! The term "Inhour" is actually a portmanteau of **In**verse **Hour**. It was an early unit of reactivity defined by Enrico Fermi's team at Chicago Pile-1 as the amount of positive reactivity that would result in a stable reactor period of exactly one hour. Lamarsh doesn't use that name at all, in fact, and simply calls it the reactor period equation.

This equation relates the **Reactivity** (ρ, what the operator changes with control rods) to the **Reactor Period** (T, how fast the power changes). Recall the reactor period T is the time required for power to change by a factor of e.

Note: In a real reactor, there are 6 distinct groups of precursors with different decay constants λ_i. The full equation is a sum:

$$\rho = \frac{l}{T} + \sum_{i=1}^{6} \frac{\beta_i}{1 + \lambda_i T}$$

3 Analyzing the Result

Let's analyze the One-Group equation to understand the three distinct regimes of reactor operation.

3.1 Regime 1: Sub-Prompt Critical (Normal Operation)

Suppose we insert a very small amount of reactivity: $\rho \ll \beta$. Assume $\rho = 0.001$ (0.1%). For U-235, $\beta \approx 0.0065$ (0.65%). Since l is tiny (10^{-4}), the first term l/T is negligible for large T. The equation simplifies to:

$$\rho \approx \frac{\beta}{1 + \lambda T}$$

Solving for Period T:

$$T \approx \frac{\beta - \rho}{\lambda \rho}$$

Using $\lambda \approx 0.1\ \text{s}^{-1}$ (average precursor decay constant):

$$T \approx \frac{0.0065 - 0.001}{(0.1)(0.001)} = \frac{0.0055}{0.0001} = \mathbf{55} \text{ seconds}$$

Result: Instead of 0.1 seconds (Prompt), the period is 55 seconds. This is slow! A human operator can easily watch a gauge and adjust a rod in 55 seconds.

3.2 Regime 2: The Threshold (Prompt Critical)

What happens if we add enough reactivity such that $\rho = \beta$?

$$\beta \approx \frac{l}{T} + \frac{\beta}{1 + \lambda T}$$

The math blows up. T must become very small for this equality to hold. Physically, this means the reactor is critical on prompt neutrons *alone*. The delayed neutrons are no longer needed to sustain the chain reaction.

3.3 Regime 3: Super-Prompt Critical (The Danger Zone)

If $\rho > \beta$, the reactor is supercritical on prompt neutrons alone. The delayed neutrons are still being produced, but the prompt chain reaction is growing so rapidly that they are essentially left behind.

For very short periods, the Inhour equation reduces to focus on the excess prompt reactivity $(\rho - \beta)$:

$$\rho \approx \frac{l}{T} + \beta \quad \Longrightarrow \quad T \approx \frac{l}{\rho - \beta}$$

Suppose we insert a reactivity just slightly above prompt critical: $\rho = \beta + 0.001 = 0.0075$. The reactor period is now dictated entirely by that tiny excess:

$$T \approx \frac{10^{-4}}{0.0075 - 0.0065} = \frac{10^{-4}}{0.001} = \mathbf{0.1} \text{ s}$$

We return to the "Bomb" kinetics of Lecture 15. Even barely crossing the threshold drops the period to fractions of a second. The reactor power will multiply by a factor of e ten times every second, making it completely uncontrollable by mechanical control rods.

3.4 Regime 4: Large Negative Reactivity (The SCRAM)

We have discussed what happens when you pull rods out, but what happens when you slam them all in to shut down the reactor (a SCRAM)?

Suppose we drop the control rods, inserting a massive negative reactivity of $\rho = -\$10.00$ (i.e., $-10 \times \beta$).

Looking at the physics of the "Storage Buffer":

1. **The Prompt Drop:** The prompt neutron population, which has a lifetime of 10^{-4} seconds, collapses instantly. The reactor power rapidly drops by a few decades in a fraction of a second.

2. **The Precursor Tail:** The power does not drop to zero! The precursors that were created before the SCRAM are still sitting in the fuel. They continue to decay, releasing delayed neutrons into the subcritical core. A nice plot of the power after SCRAM can be found at www.nuclear-power.com.

If we look at the extreme limit of the Inhour equation for large negative reactivity, the period T approaches the mean life of the longest-lived precursor group. The longest-lived precursor is Bromine-87, with a half-life of 55 seconds (which corresponds to a mean life $\tau = 55/\ln(2) \approx 80$ seconds).

Result: No matter how much negative reactivity you insert, once the prompt drop is over, the reactor power will decay on an unchangeable **-80 second period**. The reactor is chained to the physics of Bromine-87. Once the prompt population is gone, the remaining power decay rate is fundamentally limited by delayed-neutron precursor decay.

4 Effective Delayed Neutron Fraction (β_{eff})

In the Inhour equation, you might have noticed the term β_{eff} instead of just β. Why is the "effective" fraction different from the physical fraction of delayed neutrons?

It comes down to the energy of the neutrons at birth:

- **Prompt Neutrons** are born at an average energy of ~ 2.0 MeV.
- **Delayed Neutrons** are born from the decay of precursors at a much lower average energy of ~ 0.4 MeV.

Because delayed neutrons are born at lower energies, they do not have to migrate as far through the moderator to reach thermal energies. Using the leakage concepts from last week, delayed neutrons have a smaller Fermi Age (τ_T), which means they have a significantly higher **Fast Non-Leakage Probability (P_{FNL})**.

Because they are less likely to leak out of the core, a single delayed neutron is actually "worth more" to the chain reaction than a single prompt neutron! We define an Importance Factor (γ):

$$\beta_{eff} = \gamma\beta \tag{2}$$

In a large commercial Light Water Reactor, $\gamma \approx 1.04$. However, in a small, highly leaky research reactor, γ can be as high as 1.20, meaning the delayed neutrons are 20% more effective at sustaining the chain reaction than the prompt neutrons.

5 Units of Reactivity: The Dollar

Because the fraction β is the most important safety limit in a reactor, we use it as a unit of measurement.

- **1 Dollar (\$1.00)** = A reactivity of β.
- **1 Cent (1¢)** = $0.01 \times \beta$.

For U-235 ($\beta = 0.0065$):

- $\$1.00 = 0.0065\Delta k/k$
- $\$0.50 = 0.00325\Delta k/k$ (Safe)
- $\$1.10 = 0.00715\Delta k/k$ (Prompt Critical - Dangerous)

"The reactor is rising on a 50-cent period" is standard control room terminology.

5.1 Case Study: The Plutonium Problem

To understand why we use Dollars instead of just reporting absolute $\Delta k/k$, imagine two identical operators working on two different reactors. Operator A is running a standard U-235 reactor. Operator B is running an experimental Pu-239 Fast Reactor.

Both operators make the exact same minor error: they accidentally withdraw a rod that inserts an absolute reactivity of $\Delta k/k = 0.004$ (just 0.4%).

- **Operator A (U-235):** Since $\beta = 0.0065$, the reactivity is $\rho = 0.004/0.0065 = \$0.61$. Looking at the Inhour equation, this puts the reactor on a stable, easily manageable period of about 10 seconds. Operator A calmly inserts the rod back in.
- **Operator B (Pu-239):** Since $\beta = 0.0021$, the exact same rod movement yields $\rho = 0.004/0.0021 = \$1.90$. Operator B just went prompt critical, and the core melts in milliseconds.

The Lesson: Absolute reactivity ($\Delta k/k$) doesn't tell you how dangerous a reactor state is unless you know the fuel's specific β. 50 cents is identically safe in both reactors, even though the physical rod movements required to get there are vastly different.

6 Physical Insight: Why does it work?

Think of the delayed neutrons as a "storage buffer."

- When you increase reactivity, the prompt neutrons rise instantly.
- But the delayed neutrons (0.65% of the population) are "stuck" in the precursors (Bromine/Iodine atoms). They haven't been born yet.
- The total population cannot rise to the new equilibrium level until those precursors decay and release their neutrons.
- This "drag" holds the reactor back, forcing it to wait for the precursors (approx. 10 to 80 seconds).

7 Case Study: Chicago Pile-1 and the Value of Time

We now return to the first nuclear reactor, Chicago Pile-1 (CP-1), not to celebrate its success, but to understand why it *did not fail*. This case study is fundamentally about **time scales**: how fast the neutron population grows, how fast humans and machines can respond, and what happens when those two clocks no longer overlap.

CP-1 was an uncooled, graphite-moderated pile of natural uranium metal and oxide. Because neutrons diffused for long distances in the graphite before being absorbed, the prompt neutron lifetime was unusually long:

$$l \approx 10^{-3} \text{ s}$$

The delayed neutron fraction for U-235 was:

$$\beta \approx 0.0065$$

There were no engineered negative temperature coefficients in the modern sense. No coolant density feedback. No rapid shutdown mechanisms. The *only* stabilizing mechanism available during startup was the presence of delayed neutrons.

7.1 The Approach to Criticality and the Lunch Break

On December 2, 1942, CP-1 was assembled incrementally, layer by layer. At each step, Fermi's group measured the neutron multiplication and plotted $1/M$ versus the number of layers. As you saw earlier, this graph was nearly linear. Extrapolating to $1/M = 0$ told them exactly where criticality would occur.

The pile had multiple control rods, mostly made of cadmium nailed to wood. There was a set of automatic control rods, an emergency "ZIP" rod suspended by a rope, and a final regulating rod operated manually by George Weil. They pulled the rods in stages, measuring the neutron counts. Late in the morning, as the $1/M$ plot showed they were agonizingly close to criticality, an automatic safety rod suddenly dropped because its trip threshold was set too low. As the room held its breath, Fermi famously looked at the clock and declared, "I'm hungry. Let's go to lunch."

This was not just legendary coolness under pressure; it was a testament to the physics. By ensuring they were strictly subcritical, the pile was entirely stable. They could safely walk away.

7.2 What Actually Happened: A Microscopic Reactivity Insertion

After lunch, they returned for the final approach. When Weil withdrew the regulating rod for the final time, the reactivity insertion was deliberately kept microscopic. Suppose that final, very slow rod motion inserted a reactivity of only about 3 cents:

$$\rho = 0.0002$$

Because $\rho \ll \beta$, the reactor was *delayed critical*. The power rise was governed by delayed neutrons, with a reactor period:

$$T \approx \frac{\beta - \rho}{\lambda \rho} = \frac{0.0065 - 0.0002}{(0.1)(0.0002)} \approx 315 \text{ s}$$

What this meant operationally: A period of over 5 minutes is glacial by reactor standards. It meant the power was creeping up so slowly that if a control rod stuck, there was time to think, time to act, and time to recover.

- George Weil could push the rod back in manually.
- Norman Hilberry, the famous "Axe Man," could sever the rope suspending the emergency cadmium rod.
- The "Suicide Squad" on top of the pile could dump buckets of cadmium-sulfate solution into the graphite.

These were crude safety systems by modern standards, but they were *adequate*—because the physics unfolded slowly enough for humans to participate.

7.3 Alternate History: The 76-Layer Blueprint

Now consider a counterfactual scenario based on Fermi's original designs. Fermi initially calculated that CP-1 would need to be a sphere of 76 layers to reach criticality. However, the uranium metal and graphite supplied for the project ended up being much purer than anticipated. This increased k_∞, which decreased the critical geometric buckling. As a result, the pile went critical at only 57 layers, taking the shape of a flattened, oblate spheroid.

Suppose they had operated like a standard construction crew, ignoring the $1/M$ plot and blindly building the pile to the 76-layer blueprint. By layering massive amounts of fuel and moderator far beyond the critical size, they would have banked a tremendous amount of excess reactivity. If they had then fully withdrawn the control rod banks, they might have easily uncovered a reactivity of:

$$\rho = 0.0165$$

Now, $\rho > \beta$.

The reactor would be *prompt critical*. Delayed neutrons would no longer control the dynamics. The period would be set by the prompt lifetime:

$$T \approx \frac{l}{\rho - \beta} = \frac{10^{-3}}{0.010} = 0.1 \text{ s}$$

7.4 Why the Safety Systems (Even the Automatic Ones) Would Not Have Saved It

It is important to note that Fermi did have an automatic safety rod. It was held out of the pile by an electromagnet wired to a neutron counter. If the neutron flux exceeded a preset limit, a relay would trip, the magnet would de-energize, and a heavy weight would pull the rod into the core.

However, in a prompt critical scenario, even this automatic system would be useless against the physics of a 0.1-second period.

Assume the pile is operating at 1 W when the final rod clears the core. With a 0.1-second period:

$$P(t) = P_0 e^{t/T}$$

Let's break down the best-case response time of the 1942 automatic safety system:

- **Instrument Delay:** The neutron counter, relay, and electromagnet de-energization take roughly 0.1 to 0.2 seconds.
- **Gravity and Inertia:** The pile was roughly 20 feet (6 meters) tall. For the weighted rod to drop 3 meters into the high-worth center of the core requires physical travel time. From basic kinematics ($d = \frac{1}{2}at^2$), even assuming a frictionless drop at $1g$, it takes nearly 0.8 seconds for the rod to reach the center.

The absolute fastest the automatic rod could insert negative reactivity was about 1.0 second.

In that 1.0 second, the reactor power would have already increased by a factor of $e^{10} \approx 22,000$. By 1.5 seconds—the time required for the rod to fully insert—the power would be:

$$P(1.5) = (1 \text{ W})e^{1.5/0.1} = e^{15} \approx 3.2 \times 10^6 \text{ W}$$

If the automatic system was paired with human action (like Hilberry cutting the rope for the backup ZIP rod), the total response time stretches to 2 seconds or more, at which point the power exceeds hundreds of megawatts.

Crucially:

- The instantaneous Doppler feedback in CP-1's natural uranium lumps was far too weak to quench a large prompt excursion before the metal melted.
- Heating of the graphite moderator would lag far behind the neutron excursion.

- Thermal expansion would not significantly change leakage before catastrophic heating.

Once prompt criticality was reached, it is not clear that *anything* could have terminated the excursion short of fuel damage and ignition of the graphite moderator. The pile had no fast, intrinsic negative feedback capable of outrunning the prompt kinetics. An automatic rod relying on gravity and 1940s relays simply cannot catch a nuclear transient operating on a millisecond timescale.

7.5 The Real Safety Margin

The safety of CP-1 did not come from materials, geometry, or engineered systems. It came from Fermi's insistence on understanding the multiplication factor *before* approaching criticality.

By carefully extrapolating the $1/M$ curve and ensuring that $\rho < \beta$ at every step, he guaranteed that the reactor's time constant remained long compared to human response times.

The lesson: Delayed neutrons did not merely make CP-1 controllable. They made it survivable. Crossing the prompt critical threshold would have compressed the dynamics into a regime where control, feedback, and even intent no longer mattered.

8 Summary

1. **Point Kinetics:** We couple the neutron population to the precursor concentration.
2. **The Inhour Equation:** Relates Reactivity (ρ) to Period (T).
3. **Safety Limit:** We must keep $\rho < \beta$ (less than \$1.00).
4. **Normal Operation:** As long as $\rho < \beta$, the reactor period is determined by the slow decay of precursors (λ), making the reactor controllable by humans.
5. **General Reference:** A very accessible reference for reactor kinetics is the DOE Handbook Nuclear Physics and Reactor Theory. I've found it a bit easier to read than Lamarsh (which it draws heavily from).

A Derivation of the Inhour Equation

This appendix provides the step-by-step algebraic derivation of the one-group Inhour equation starting from the point kinetics equations introduced in Section 2.

Step 1: The Point Kinetics Equations

Recall our two coupled differential equations for the neutron population (n) and the delayed neutron precursor concentration (C):

$$\frac{dn}{dt} = \frac{k(1-\beta)n}{l} + \lambda C - \frac{n}{l} \tag{3}$$

$$\frac{dC}{dt} = \frac{k\beta n}{l} - \lambda C \tag{4}$$

We want to find the asymptotic solution where both the neutron population and the precursor concentration grow (or decay) exponentially with the same constant reactor period, T. Therefore, we assume solutions of the form:

$$n(t) = n_0 e^{t/T} \tag{5}$$

$$C(t) = C_0 e^{t/T} \tag{6}$$

Step 2: Solve the Precursor Equation

Substitute our assumed exponential forms into the precursor equation (4):

$$\frac{d}{dt}\left(C_0 e^{t/T}\right) = \frac{k\beta}{l}\left(n_0 e^{t/T}\right) - \lambda\left(C_0 e^{t/T}\right)$$

Taking the derivative gives:

$$\frac{C_0}{T}e^{t/T} = \frac{k\beta n_0}{l}e^{t/T} - \lambda C_0 e^{t/T}$$

Divide the entire equation by $e^{t/T}$:

$$\frac{C_0}{T} = \frac{k\beta n_0}{l} - \lambda C_0$$

Now, isolate C_0 by gathering terms:

$$C_0\left(\frac{1}{T} + \lambda\right) = \frac{k\beta n_0}{l}$$

$$C_0 = \frac{k\beta n_0}{l\left(\frac{1}{T} + \lambda\right)} = \frac{k\beta T n_0}{l(1+\lambda T)}$$

Step 3: Substitute into the Neutron Equation

Now, substitute our exponential forms and our new expression for C_0 into the neutron equation (3):

$$\frac{n_0}{T}e^{t/T} = \frac{k(1-\beta)n_0}{l}e^{t/T} + \lambda C_0 e^{t/T} - \frac{n_0}{l}e^{t/T}$$

Divide by $n_0 e^{t/T}$:

$$\frac{1}{T} = \frac{k(1-\beta)-1}{l} + \lambda\frac{C_0}{n_0}$$

Substitute the ratio $\frac{C_0}{n_0}$ from our precursor solution:

$$\frac{1}{T} = \frac{k(1-\beta)-1}{l} + \lambda\left[\frac{k\beta T}{l(1+\lambda T)}\right]$$

Step 4: Algebraic Manipulation to Find Reactivity (ρ)

Multiply the entire equation by l to clear the denominators:

$$\frac{l}{T} = k(1-\beta) - 1 + \frac{\lambda k\beta T}{1+\lambda T}$$

Next, divide the entire equation by k:

$$\frac{l}{kT} = 1 - \beta - \frac{1}{k} + \frac{\lambda\beta T}{1+\lambda T}$$

Recall the definition of reactivity, $\rho = \frac{k-1}{k} = 1 - \frac{1}{k}$. We can group the $1 - \frac{1}{k}$ terms to substitute ρ:

$$\frac{l}{kT} = \rho - \beta + \frac{\lambda\beta T}{1+\lambda T}$$

Now, rearrange the equation to solve for ρ:

$$\rho = \frac{l}{kT} + \beta - \frac{\lambda\beta T}{1+\lambda T}$$

Factor out β from the last two terms:

$$\rho = \frac{l}{kT} + \beta\left(1 - \frac{\lambda T}{1+\lambda T}\right)$$

Find a common denominator for the terms inside the parenthesis:

$$\rho = \frac{l}{kT} + \beta\left(\frac{1+\lambda T-\lambda T}{1+\lambda T}\right)$$

The λT terms cancel out, leaving the final one-group Inhour equation (where k is the effective multiplication factor k_{eff}):

$$\boxed{\rho = \frac{l}{Tk_{eff}} + \frac{\beta}{1+\lambda T}} \qquad (7)$$

Extension to Six Groups

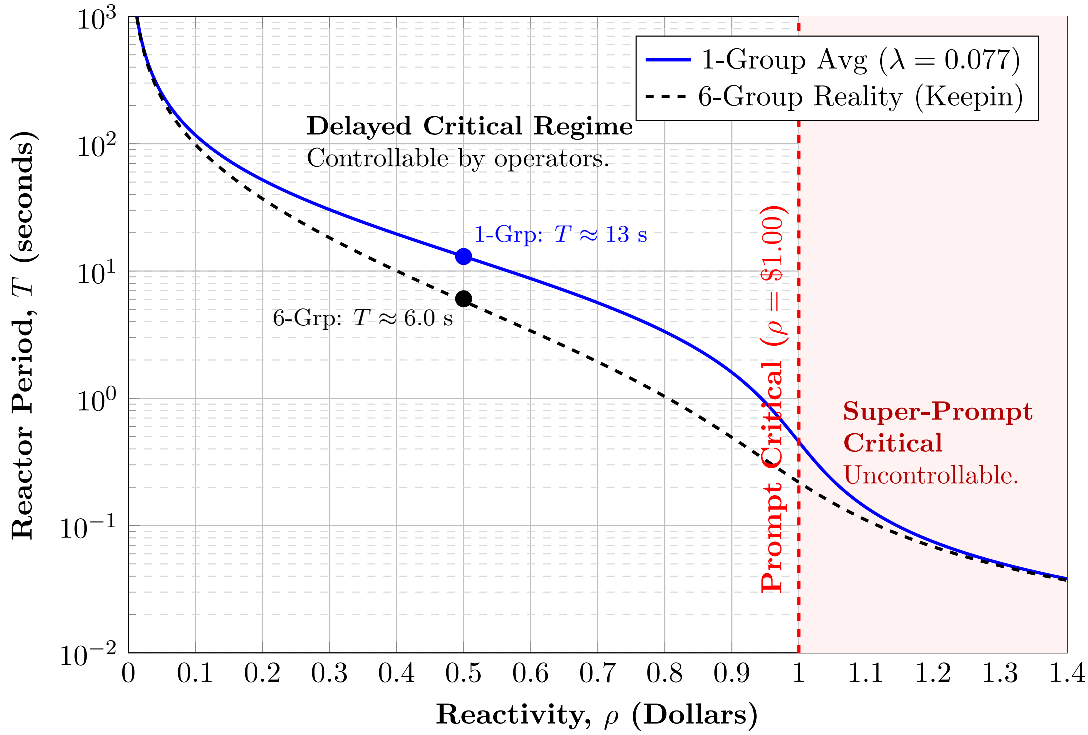
If we do not make the one-group approximation, we must write a separate precursor equation for each of the $i = 1$ to 6 delayed neutron groups, each with its own yield fraction (β_i) and decay constant (λ_i).

Following the exact same mathematical procedure, the $\frac{\beta}{1+\lambda T}$ term simply becomes a summation over all six groups, yielding the full Inhour Equation:

$$\rho = \frac{l}{Tk_{eff}} + \sum_{i=1}^{6} \frac{\beta_i}{1 + \lambda_i T}$$

The Inhour Curve (1-Group vs. 6-Group)

Comparison of Reactivity (ρ) vs. Reactor Period (T)



The Danger of the Average:
Notice that the 1-Group model (blue line) predicts a much slower, "lazier" reactor period than reality (black dashed line).

- The 1-Group model assumes all precursors decay at an average rate ($\lambda_{avg} \approx 0.08 s^{-1}$).
- The 6-Group reality includes "fast" precursors (Groups 5 & 6) with lifetimes of < 1 second. These decay almost immediately, contributing neutrons *now* rather than *later*.

This makes the real reactor "twitchier" (shorter period) than the simple model predicts.

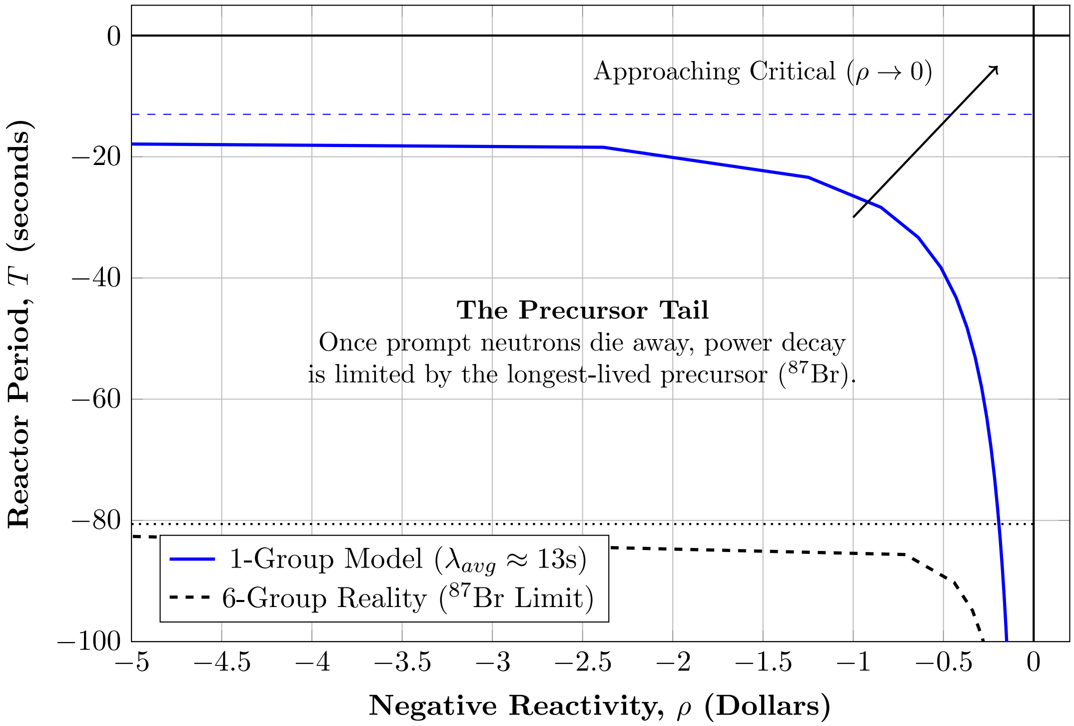
Safety Note: A larger β_{eff} is safer!

- High β_{eff} (e.g., ^{235}U, $\beta \approx 0.0065$): The gap between delayed and prompt critical is large (650 pcm).
- Low β_{eff} (e.g., ^{239}Pu, $\beta \approx 0.0021$): The gap is small (210 pcm).

In a low-β system, a much smaller physical reactivity insertion (smaller rod movement) is required to reach the prompt critical danger zone.

The Inhour Curve (Negative Reactivity / Shutdown)

Demonstrating the "Scram Limit" and the failure of the 1-Group approximation.



The Physical Meaning of the Asymptote: Unlike the positive reactivity side where the period crashes to zero, inserting massive amounts of negative reactivity results in a horizontal asymptote.

- After the initial instantaneous "Prompt Drop" in power, the reactor is sustained entirely by delayed neutrons.
- You cannot force the precursors to decay faster. The power decay rate becomes limited by the mean life of the longest-lived precursor group (Group 1, ^{87}Br, $\tau \approx 80s$).
- Therefore, the stable reactor period can never be faster than **−80 seconds**, regardless of how many control rods are inserted.

Note the massive failure of the 1-Group approximation here, which predicts a much faster shutdown period of −13 seconds because it averages out the long-lived precursors. The reactor **power** decays far more slowly (days) because of the decay heat released over time by the fission products.